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Inferring Volatility Dynamics and Risk Premia from the S&P 500 and VIX Markets

Contents

1	Big picture	3
2	Data and measurement (market setting and key objects)	4
2.1	Why the S&P 500–VIX setting is useful	4
2.2	Sample and option data	4
2.3	Option filtering and measurement choices	4
2.4	Main observed variables	5
2.5	Descriptive facts	5
2.6	Key latent objects	5
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Full affine jump-diffusion model: SVJ3	6
3.2	Jump structure and leverage	6
3.3	Benchmark model	7
3.4	Risk-premium specification	7
3.5	Integrated variance risk premium	7
3.6	VIX squared as an affine function	8
3.7	Derivative pricing	8
3.8	Particle-filter estimation	8
3.9	Pricing-error metric	9
3.10	Return-predictability regression	9
4	Identification and empirical design (what makes the estimates credible)	9
4.1	What is being identified	9
4.2	Why the “with versus without VIX options” comparison is informative	9
4.3	Why out-of-sample pricing matters	10
4.4	Why synthetic VIX indices are a useful validation	10
4.5	Model-selection credibility	10
4.6	Measurement and filtering credibility	11
4.7	Economic-application credibility	11
4.8	Remaining limitations	11

5	Tables and figures (with extracted exhibits)	11
5.1	Table 1: SVJ3 parameter estimates	11
5.2	Figure 1: S&P 500/VIX dynamics and option-implied volatility smiles	12
5.3	Table 2: Benchmark-model parameter estimates	12
5.4	Table 3: SVJ3 option-pricing errors	13
5.5	Table 4: Benchmark-model option-pricing errors	13
5.6	Table 5: Return-predictability regressions	14
5.7	Table 6: Descriptive statistics	14
5.8	Table 7: Model selection	14
5.9	Figure 2: Filtered volatility and jump trajectories	15
5.10	Table 8: Diebold–Mariano pricing-error tests	15
5.11	Figure 3: Dynamics and decomposition of the variance risk premium	16
5.12	Figure 4: Model-implied VRP versus model-free approximations	16
5.13	Figure 5: Slope and components of the VRP term structure	17
5.14	Figure 6: VRP term structure and higher-order shape	17
5.15	Figure 7: Forward-variance trading strategy	18
5.16	Table 9: Synthetic squared-VIX RMSEs	18
6	Short summary	19
7	Conclusion	19
7.1	Core conclusion	19
7.2	Economic intuition	20
7.3	Methodological takeaway	20
7.4	Asset-pricing takeaway	20
7.5	Final takeaway	20

1 Big picture

- **Question.** What additional information do VIX options contain about S&P 500 return volatility dynamics, variance risk premia, and future returns beyond the information already contained in the S&P 500 index, the VIX index, and S&P 500 options?
- **Setting.** The paper jointly studies the S&P 500 and VIX derivatives markets from **March 1, 2006 to April 30, 2016**. The data include weekly S&P 500 options, VIX options, S&P 500 index returns, VIX levels, and realized variance.
- **Core empirical problem.** S&P 500 options and VIX options are linked to the same underlying market, but they reveal different objects. S&P 500 options reveal the risk-neutral distribution of future index returns; VIX options reveal the risk-neutral distribution of future volatility. The paper asks whether the second object is spanned by the first.
- **Main contribution.** The paper estimates a flexible affine jump-diffusion model using a large, unbalanced panel of S&P 500 and VIX option prices. Unlike a one-day calibration, the estimation is time-series consistent and uses a particle filter to infer latent volatility factors and jumps.
- **Model idea.** S&P 500 returns are driven by a short-run variance factor v_t , a more persistent central-tendency variance factor m_t , and a jump-intensity factor u_t . Returns and variance can jump, and negative return jumps can coincide with positive variance jumps.
- **Empirical comparison.** The paper estimates the model twice: once *without* VIX options and once *with* VIX options. If VIX options are spanned by the S&P 500 market, adding them should not materially improve the pricing of VIX options, the variance term structure, or economic applications.
- **Main result.** VIX options contain information that is **not spanned** by S&P 500 options. In the SVJ3 model, including VIX options reduces the in-sample VIX-option RMSRE from **0.611 to 0.488** and the out-of-sample VIX-option RMSRE from **0.555 to 0.475**.
- **Model-selection result.** The richest affine specification, SVJ3, dominates nested alternatives by likelihood, AIC, and BIC. The model needs two variance factors, jumps, and a jump-intensity factor to jointly fit S&P 500 and VIX derivatives.
- **Variance-term-structure result.** Including VIX options improves the model's ability to reproduce synthetic VIX-style indices at maturities of two, three, five, and six months. For the SVJ3 model, the overall six-month squared-VIX RMSE falls from **0.031 to 0.025**.
- **Variance risk premium result.** The variance risk premium is negative. It becomes sharply more negative during market stress, especially around the 2008 crisis. The term structure is downward sloping in calm periods but becomes nonmonotonic in crises.
- **Economic application.** The slope of the VRP term structure acts as a trading signal for forward variance. A strategy that switches position when the VRP slope changes sign reaches a Sharpe ratio of **0.77**, compared with **0.01** for the basic long-forward-variance strategy.
- **Return predictability result.** Model-implied VRP variables improve forecasts of S&P 500 returns. The level of the VRP and the slope of its term structure jointly matter because they span information in the latent variance factors.

- **Economic takeaway.** VIX options are not merely redundant derivatives on the S&P 500. They reveal risk-neutral beliefs about future volatility distributions and help identify variance risk premia that matter for pricing, risk management, volatility trading, and return forecasts.

2 Data and measurement (market setting and key objects)

2.1 Why the S&P 500–VIX setting is useful

- The S&P 500 index, S&P 500 options, the VIX index, and VIX options are all tied to the same underlying equity market.
- The VIX index approximates the risk-neutral expected future realized volatility of S&P 500 returns over the next 30 days.
- A VIX option expiring at time T is linked to volatility over the window after T , while an S&P 500 option expiring at T is linked to the index level at T .
- This difference lets the paper ask whether VIX options reveal a whole distribution of future volatility that is not already embedded in S&P 500 options.

Interpretation. The markets are economically connected but not informationally identical. The key question is whether the VIX option market carries independent information about volatility dynamics.

2.2 Sample and option data

- VIX options began trading in 2006. The sample runs from **March 2006 to April 2016**.
- Option prices are weekly closing prices, mainly using Wednesday observations, from Option-Metrics.
- The in-sample estimation period is **March 2006 to February 2009**; the out-of-sample period is **March 2009 to April 2016**.
- The final data set contains **365,507** OTM S&P 500 options and **44,539** VIX call options.
- The daily average is approximately **639** S&P 500 options and **78** VIX options.

Interpretation. The paper uses a large option panel rather than a small set of near-the-money contracts. This is important because tails and maturity slices are central for identifying variance dynamics.

2.3 Option filtering and measurement choices

- The paper keeps options with maturities between one week and one year.

- It focuses on liquid OTM S&P 500 options and liquid VIX calls.
- Futures prices are inferred from put–call parity using liquid near-the-money contracts, avoiding dividend forecasts and timing mismatches.
- Implied volatilities are calculated using futures prices as the underlying.

Interpretation. The cleaning steps are designed to make option prices comparable across markets and to avoid treating illiquid or mechanically inconsistent quotes as structural information.

2.4 Main observed variables

- **S&P 500 futures returns:** observed daily log returns of the equity index futures.
- **VIX levels:** observed index levels that summarize near-term risk-neutral expected variance.
- **Option prices:** S&P 500 and VIX option prices across strikes and maturities.
- **Realized variance:** daily realized variance from high-frequency S&P 500 data, used as an additional measurement equation for the latent variance process.

Interpretation. The estimation combines spot data, derivative prices, and realized variance to identify both historical and risk-neutral volatility dynamics.

2.5 Descriptive facts

- S&P 500 returns have high kurtosis, especially during the in-sample crisis window.
- VIX returns are positively skewed and highly volatile.
- S&P 500 returns and VIX changes are strongly negatively related, motivating the use of VIX derivatives for hedging equity risk.
- S&P 500 implied volatility smiles are generally downward sloping in moneyness, while VIX implied volatility smiles tend to slope upward.

Interpretation. The two option markets reflect different tail concerns: investors buy S&P 500 puts to insure against equity crashes and buy VIX calls to insure against volatility spikes.

2.6 Key latent objects

- v_t : short-run diffusive variance of S&P 500 returns.
- m_t : persistent central tendency of variance, around which v_t mean-reverts.
- u_t : jump-intensity factor, especially useful for modeling negative return jumps and positive variance jumps.
- J_t : jump components in returns and variance.

- $\text{VRP}(t, T)$: integrated variance risk premium over horizon $[t, T]$.

Interpretation. The model separates short-run volatility spikes, persistent volatility levels, and jump risk. This separation is crucial for interpreting the VRP term structure.

3 Models and empirical specifications (all equations + interpretation)

3.1 Full affine jump-diffusion model: SVJ3

The paper models the log forward price of the S&P 500 as $Y_t = \log F_t$. Under the risk-neutral measure $\mathbb{Q}$, the main model is

$$dY_t = \mu(v_{t-}, m_t, u_t)dt + \sqrt{v_{t-}} dW_t^Y + dJ_t^{Y(+)} + dJ_t^{Y(-)}, \quad (1)$$

$$dv_t = \kappa_v(m_t - v_{t-})dt + \sigma_v \sqrt{v_{t-}} dW_t^v + dJ_t^{v(+)} + dJ_t^{v(-)}, \quad (2)$$

$$dm_t = \kappa_m(\theta_m - m_t)dt + \sigma_m \sqrt{m_t} dW_t^m, \quad (3)$$

$$du_t = \kappa_u(\theta_u - u_t)dt + \sigma_u \sqrt{u_t} dW_t^u. \quad (4)$$

Interpretation.

- Y_t is the return process.
- v_t is the current diffusive variance.
- m_t is a slow-moving variance level; it captures long-run volatility conditions.
- u_t affects the intensity of negative jumps in returns.
- The model is affine, so characteristic functions and VIX-related claims can be priced efficiently.

3.2 Jump structure and leverage

The jump processes are driven by positive and negative return-jump counting processes:

$$J_t^{Y(+)} = \sum_{i=1}^{N_t^{Y(+)}} Z_i^{Y(+)}, \quad J_t^{Y(-)} = \sum_{i=1}^{N_t^{Y(-)}} Z_i^{Y(-)}, \quad (5)$$

$$J_t^{v(+)} = \sum_{i=1}^{N_t^{Y(+)}} Z_i^{v(+)}, \quad J_t^{v(-)} = \sum_{i=1}^{N_t^{Y(-)}} Z_i^{v(-)}. \quad (6)$$

The jump intensities are affine in the latent state vector $X_t = (v_t, m_t, u_t)'$:

$$\lambda^{(+)}(v_{t-}, m_t) = \lambda_0^{(+)} + \lambda^{(+)' } X_{t-}, \quad (7)$$

$$\lambda^{(-)}(v_{t-}, m_t, u_t) = \lambda_0^{(-)} + \lambda^{(-)' } X_{t-}. \quad (8)$$

Interpretation. Negative return jumps and positive variance jumps are allowed to occur together. This captures the leverage-style fact that equity crashes tend to coincide with volatility spikes.

3.3 Benchmark model

The paper also estimates a simpler two-factor benchmark model:

$$dY_t = \mu(v_{t-}, m_t)dt + \sqrt{v_{t-}} dW_t^Y + dJ_t^Y, \quad (9)$$

$$dv_t = \kappa_v(m_t - v_{t-})dt + \sigma_v \sqrt{v_{t-}} dW_t^v + dJ_t^v, \quad (10)$$

$$dm_t = \kappa_m(\theta_m - m_t)dt + \sigma_m \sqrt{m_t} dW_t^m. \quad (11)$$

Interpretation. The benchmark keeps a two-factor stochastic volatility structure but removes the separate jump-intensity factor and negative jumps in variance. It is useful for showing which results are robust to a simpler model.

3.4 Risk-premium specification

The change from the risk-neutral measure $\mathbb{Q}$ to the historical measure $\mathbb{P}$ preserves the affine structure. The equity risk premium contains a diffusive component and a jump component. In simplified notation,

$$\gamma_t = \eta_Y v_{t-} + \text{compensation for positive jump risk} + \text{compensation for negative jump risk}. \quad (12)$$

Interpretation. Investors are compensated not only for continuous equity risk but also for jump risk. Because jumps affect volatility, the jump-risk component is important for the variance risk premium.

3.5 Integrated variance risk premium

For horizon $[t, T]$, the integrated variance risk premium is defined as the difference between expected integrated variance under the historical and risk-neutral measures:

$$\text{VRP}(t, T) = \frac{1}{T-t} \left(\mathbb{E}_t^{\mathbb{P}} \left[\int_t^T v_s ds + \text{jump-variance terms} \right] - \mathbb{E}_t^{\mathbb{Q}} \left[\int_t^T v_s ds + \text{jump-variance terms} \right] \right). \quad (13)$$

Interpretation. A negative VRP means that risk-neutral expected variance is higher than physical expected variance. Investors are willing to pay for protection against variance shocks.

3.6 VIX squared as an affine function

Because of the affine model structure, the model-implied squared VIX can be written as an affine function of the latent factors:

$$\text{VIX}_t^2(\tau) = A(\tau) + B(\tau)v_t + D(\tau)m_t + G(\tau)u_t. \quad (14)$$

Interpretation. VIX options are informative because they price claims on a future affine function of the latent variance state. They therefore help identify the distribution of future volatility, not just its current level.

3.7 Derivative pricing

Option prices are computed under $\mathbb{Q}$. For a generic option payoff $g(\cdot)$, the model price is

$$P_t = e^{-r(T-t)} \mathbb{E}_t^{\mathbb{Q}}[g(\cdot)]. \quad (15)$$

For VIX options, the payoff depends on VIX_T , which is a nonlinear function of VIX_T^2 . The paper uses Fourier-cosine methods and affine characteristic functions to price S&P 500 and VIX options efficiently.

Interpretation. Tractability matters because the data set contains hundreds of options per date. A model that cannot be priced quickly cannot be estimated on the full panel.

3.8 Particle-filter estimation

The unobserved state includes latent factors and jump information:

$$L_t = ig(v_t, m_t, u_t, \Delta N_t^{(+)}, \Delta N_t^{(-)}, Z_t^{Y(+)}, Z_t^{Y(-)}, Z_t^{v(+)}, Z_t^{v(-)}). \quad (16)$$

The filter approximates

$$p(L_t \mid y^t), \quad (17)$$

where y^t denotes all observed data up to time t : index returns, VIX levels, realized variance, and option prices.

Interpretation. The auxiliary particle filter turns a nonlinear latent-state option-pricing model into an estimable time-series model. It lets the paper infer latent volatility trajectories and estimate $\mathbb{P}$ - and $\mathbb{Q}$ -parameters jointly.

3.9 Pricing-error metric

The main pricing-error statistic is the root mean squared relative error:

$$\text{RMSRE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{P_i^{\text{model}} - P_i^{\text{market}}}{P_i^{\text{market}}} \right)^2}. \quad (18)$$

Interpretation. RMSRE emphasizes proportional errors. This is important because option prices vary substantially across strikes, maturities, and underlying markets.

3.10 Return-predictability regression

For horizon h , the paper estimates regressions of the form

$$r_{t,t+h} = \alpha + \beta' X_t + \varepsilon_{t+h}, \quad (19)$$

where X_t can include past realized variance, model-implied one-month VRP, the slope of the VRP term structure, the model-free $RV - VIX^2$ proxy, and latent factors v_t, m_t, u_t .

Interpretation. If the VRP is a priced risk premium, it should forecast future returns. The slope of the VRP term structure matters because the level alone does not fully span the latent variance state.

4 Identification and empirical design (what makes the estimates credible)

4.1 What is being identified

The paper is not estimating a policy treatment effect. It is identifying whether VIX options contain *incremental information* about volatility dynamics that is not spanned by the S&P 500 market.

Main identifying comparison:

$$\text{Model estimated without VIX options} \quad \text{versus} \quad \text{Model estimated with VIX options}. \quad (20)$$

Interpretation. If the two estimations produce the same VIX-option prices, variance term structures, and economic implications, VIX options would be redundant. They do not.

4.2 Why the “with versus without VIX options” comparison is informative

- Both estimations use S&P 500 returns, VIX levels, and S&P 500 options.

- The only difference is whether VIX options are included in the estimation data set.
- VIX options strongly improve the pricing of VIX options and the synthesized variance term structure.
- The improvement holds in sample and out of sample.

Interpretation. The comparison isolates the information contribution of VIX options relative to the S&P 500 market.

4.3 Why out-of-sample pricing matters

- A purely flexible model could overfit in-sample VIX options.
- The paper tests whether the estimated parameters continue to price later VIX options after the in-sample period.
- The out-of-sample VIX-option RMSRE improves from **0.555 to 0.475** in the SVJ3 model when VIX options are included in the estimation.

Interpretation. The evidence is not just mechanical overfitting. VIX options help identify more stable volatility dynamics.

4.4 Why synthetic VIX indices are a useful validation

- The VIX index measures 30-day expected variance.
- The paper constructs VIX-style indices at longer maturities using S&P 500 options.
- These synthetic indices are not VIX options themselves, so they provide a separate check on the variance term structure.

Interpretation. If the model estimates the volatility distribution correctly, it should reproduce variance claims at maturities not directly used in the VIX index.

4.5 Model-selection credibility

- The paper estimates nested model restrictions.
- Removing jumps, reducing the number of variance factors, or eliminating the jump-intensity factor worsens likelihood and information criteria.
- SVJ3 is preferred under both estimation data sets.

Interpretation. The main results are not based on an arbitrary model choice. The data favor the richer specification.

4.6 Measurement and filtering credibility

- Option prices and VIX levels are observed with error, which accounts for bid–ask spreads, timing errors, and VIX-index construction errors.
- Realized variance enters as an additional measurement equation, helping pin down the physical variance state.
- The auxiliary particle filter is designed to handle nonlinear measurements and jumps.

Interpretation. The paper acknowledges that market prices are noisy and estimates latent states probabilistically rather than treating each price as exact.

4.7 Economic-application credibility

- The estimated VRP term structure produces trading signals with large performance differences during crisis weeks.
- The estimated VRP variables improve return-predictability regressions.
- These applications use the model-implied objects outside the immediate pricing-error exercise.

Interpretation. The model is useful not only for fitting option prices but also for economically meaningful volatility and return applications.

4.8 Remaining limitations

- The approach is parametric, so misspecification is possible.
- VIX option data are mostly call-heavy and especially informative about the right tail of volatility.
- Overlapping return windows in predictability regressions require caution when interpreting long-horizon results.
- The strategy evidence is suggestive and depends on the existence and implementation of tradable forward-variance positions.

Interpretation. The paper provides strong evidence that VIX options add information, but the exact economic magnitudes depend on the maintained affine model and market-implementation assumptions.

5 Tables and figures (with extracted exhibits)

5.1 Table 1: SVJ3 parameter estimates

Read:

- Table 1 estimates the full SVJ3 model, with and without VIX options.
- The model includes two variance factors, return and variance jumps, and a jump-intensity factor.
- The likelihood and information criteria favor the full specification relative to simpler models reported later.

Economic message. The volatility process needs both persistent and short-run components, and jump risk is quantitatively important.

Table 1

Parameters of the SVJ3 model dynamics are given in Eqs. (1)–(3). The standard errors are in italics below each parameter, except for θ_m^0 , which is calculated as $\theta_m^0 = \theta_m^0 \kappa_m^0 / \kappa_m^0$. The estimation results in the columns “w/o” are based on S&P 500 and VIX index levels and S&P 500 options; those in the columns “w/” additionally use VIX options. The LL, AIC, and BIC test values are reported below the parameters. The estimation period is from March 2006 to February 2009.

	P - and Q -parameters		P -parameters				Q -parameters	
	w/o	w/	w/o	w/			w/o	w/
$\lambda_1^{(-)}$	1.60	1.95	κ_p^0	7.40	7.30	κ_v^0	5.60	5.00
	0.22	0.21		0.22	0.12		0.21	0.18
$\lambda_2^{(-)}$	1.60	1.90	κ_m^0	0.13	0.24	κ_m^0	0.24	0.08
	0.77	1.22		0.01	0.01		0.04	0.02
$\lambda_3^{(-)}$	0.08	0.08	v_p^0	0.02	0.03	θ_m^0	0.02	0.04
	0.03	0.05		0.01	0.02		0.01	0.01
$\lambda_4^{(+)}$	0.03	0.09	$\mu_V^{(-)}$	-0.04	-0.03	$v_V^{(+)}$	0.05	0.07
	0.02	0.09		0.01	0.02		0.04	0.01
$\lambda_5^{(+)}$	1.30	1.02	$\mu_V^{(+)}$	0.02	0.02	$\mu_V^{(-)}$	-0.11	-0.11
	0.16	0.82		0.01	0.01		0.01	0.01
σ_m	0.10	0.14	$\mu_V^{(+)}$	0.60	0.58	$\mu_V^{(+)}$	0.02	0.04
	0.03	0.01		0.14	0.05		0.02	0.01
σ_v	0.65	0.69				LL	10,473	9949
	0.02	0.02				AIC	-20,898	-19,850
σ_u	0.35	0.30				BIC	-20,787	-19,739
	0.01	0.03						
ρ_{Vv}	-0.78	-0.74						
	0.01	0.02						
κ_u	2.50	4.00						
	0.27	1.12						
α_0	0.30	0.31						
	0.01	0.03						
α_1	0.65	0.65						
	0.02	0.03						
$v_V^{(-)}$	-0.04	-0.03						
	0.01	0.01						

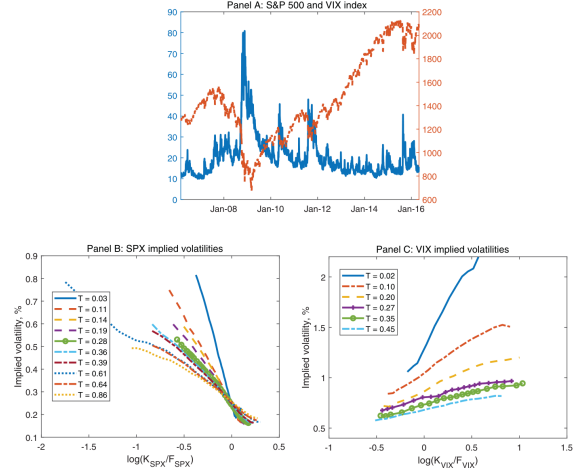


Fig. 1. This figure illustrates some characteristics of the S&P 500 and VIX markets. Panel A plots the times series of S&P 500 (dashed curve) and VIX (solid curve) indexes from March 1, 2006, to April 30, 2016. Panels B and C represent the implied volatilities of S&P 500 options and VIX options on May 10, 2010, as a function of log-moneyness. The maturities T are quoted in years.

5.2 Figure 1: S&P 500/VIX dynamics and option-implied volatility smiles

Read:

- Panel A shows the joint evolution of the S&P 500 and VIX from 2006 to 2016.
- Panels B and C compare S&P 500 and VIX implied-volatility smiles.
- S&P 500 implied volatility tends to be downward sloping, while VIX implied volatility tends to be upward sloping.

Economic message. S&P 500 options price crash insurance; VIX options price volatility-spike insurance. These are related but distinct risk objects.

5.3 Table 2: Benchmark-model parameter estimates

Read:

- Table 2 estimates the simpler two-factor benchmark model.

- The benchmark remains useful for robustness, but it has lower likelihood and worse information criteria than SVJ3.

Economic message. Two-factor stochastic volatility captures much of the data, but the richer SVJ3 model is needed to fit the joint S&P 500–VIX system better.

Table 2

Parameters of the benchmark model dynamics given in Eq. (11). The standard errors are in italics below each parameter, except for θ_m^p , which is calculated as $\theta_m^p = \theta_m^p \kappa_m^p / \kappa_m^p$. The estimation results in the columns “w/o” are based on S&P 500 and VIX index levels and S&P 500 options; those in the columns “w/” additionally use VIX options. The estimation period is from March 2006 to February 2009.

	P- and Q-parameters		P-parameters		Q-parameters			
	w/o	w/	w/o	w/	w/o	w/		
λ_0^p	0.11	0.03	κ_p^p	7.09	7.03	κ_m^p	5.36	5.34
	0.05	0.04		0.20	0.18		0.15	0.19
λ_1^p	1.20	0.65	κ_m^p	0.19	0.27	κ_m^Q	0.17	0.21
	0.22	0.31		0.04	0.05		0.09	0.09
λ_2^p	3.08	3.32	ν_p^p	0.02	0.01	θ_m^Q	0.04	0.05
	1.22	1.19		0.00	0.02		0.01	0.01
σ_m	0.20	0.21	μ_V^p	0.00	-0.01	ν_V^Q	0.04	0.08
	0.05	0.02		0.00	0.00		0.01	0.02
σ_v	0.64	0.78	σ_V^p	0.03	0.04	μ_V^Q	-0.08	-0.08
	0.03	0.02		0.01	0.02		0.02	0.02
ρ_{Vv}	-0.78	-0.78	η_V	0.45	0.25	σ_V^Q	0.12	0.12
	0.02	0.02		0.30	0.21		0.03	0.02
α_0	0.30	0.35			LL	10,370	9538	
	0.10	0.14			AIC	-20,704	-19,040	
α_1	0.60	0.55			BIC	-20,621	-18,957	
	0.04	0.03						

Table 3

RMSRE on the implied volatilities of S&P 500 options (Panel A) and VIX options (Panel B) using the SVJ3 model dynamics given in Eqs. (1)–(3) with different estimation data sets. The columns “w/o” use S&P 500 and VIX index levels and S&P 500 options; “w/” additionally use VIX options. Option pricing errors are reported conditional on moneyness (Mon.) and time to maturity (TTM) in months (M). The last column reports the number of options (#) in each category. Moneyness is defined as the ratio between the strike and the corresponding maturity future’s price.

Panel A: RMSRE on implied volatilities of S&P 500 options									
	In-sample			In-sample until Sep-08			Out-of-sample		
	w/o	w/	#	w/o	w/	#	w/o	w/	#
Overall	0.096	0.099	15952	0.073	0.079	13077	0.146	0.122	76,540
Mon. < 0.7	0.154	0.125	1776	0.079	0.049	1147	0.962	0.963	15,651
0.7 ≤ Mon. ≤ 0.95	0.091	0.068	6969	0.080	0.055	6098	0.136	0.107	31,703
0.95 ≤ Mon. ≤ 1.05	0.079	0.095	3848	0.076	0.094	3497	0.097	0.093	15,646
1.05 ≤ Mon. ≤ 1.2	0.019	0.019	2557	0.017	0.017	2048	0.136	0.126	11,062
Mon. > 1.2	0.110	0.116	802	0.050	0.092	287	0.118	0.139	2478
TTM ≤ 2 M	0.105	0.106	4997	0.085	0.092	4087	0.134	0.113	27,615
2 M < TTM < 6 M	0.088	0.078	6069	0.068	0.061	5004	0.144	0.116	28,618
TTM > 6 M	0.106	0.083	4886	0.085	0.062	3986	0.164	0.142	20,307
Panel B: RMSRE on implied volatilities of VIX options									
	In-sample			In-sample until Sep-08			Out-of-sample		
	w/o	w/	#	w/o	w/	#	w/o	w/	#
Overall	0.611	0.488	4768	0.555	0.372	3562	0.555	0.475	39,612
Mon. > 1.3	0.364	0.286	1312	0.095	0.056	4680	0.408	0.322	18,952
1.1 ≤ Mon. ≤ 1.3	0.386	0.272	1055	0.105	0.057	3120	0.426	0.314	6341
0.9 ≤ Mon. ≤ 1.1	0.567	0.380	1205	0.507	0.236	965	0.573	0.448	7325
Mon. < 0.9	0.942	0.806	1196	0.917	0.691	802	0.887	0.835	6994
TTM ≤ 2 M	0.624	0.526	2283	0.561	0.403	1662	0.536	0.440	6994
TTM > 2 M	0.582	0.415	2485	0.519	0.289	1900	0.563	0.489	11,355

5.4 Table 3: SVJ3 option-pricing errors

Read:

- For S&P 500 options, adding VIX options has only modest effects in sample and improves out-of-sample errors.
- For VIX options, adding VIX options sharply reduces RMSREs.
- The overall VIX-option RMSRE falls from **0.611 to 0.488** in sample and from **0.555 to 0.475** out of sample.

Economic message. VIX options are not well priced when the model is estimated only from the S&P 500 market. This is direct evidence of unspanned volatility information.

5.5 Table 4: Benchmark-model option-pricing errors

Read:

- The same pricing-error comparison is repeated under the benchmark model.
- VIX-option errors again fall when VIX options are included.
- The result is therefore not unique to the SVJ3 specification.

Economic message. The conclusion that VIX options add information is robust to a simpler volatility model.

Table 4

RMSRE on the implied volatilities of S&P 500 options (Panel A) and VIX options (Panel B) using the benchmark model dynamics given in Eq. (11) with different estimation data sets. The columns "w/o" use S&P 500 and VIX index levels and S&P 500 options; "w/" additionally use VIX options. Option pricing errors are reported conditional on moneyness (Mon.) and time to maturity (TTM) in months (M). The last column reports the number of options (#) in each category. Moneyness is defined as the ratio between the strike and the corresponding maturity future's price.

Panel A: RMSRE on implied volatilities of S&P 500 options									
	In-sample			In-sample until Sep-08			Out-of-sample		
	w/o	w/	#	w/o	w/	#	w/o	w/	#
Overall	0.112	0.117	15,952	0.086	0.089	13,077	0.143	0.119	76,540
Mon. < 0.7	0.167	0.142	1776	0.080	0.046	1147	0.962	0.962	15,651
0.7 ≤ Mon. ≤ 0.95	0.094	0.077	6969	0.073	0.053	6098	0.130	0.107	31,703
0.95 ≤ Mon. ≤ 1.05	0.112	0.137	3848	0.106	0.136	3497	0.108	0.102	15,646
1.05 ≤ Mon. ≤ 1.2	0.016	0.016	2557	0.013	0.013	2048	0.085	0.079	11,062
Mon. > 1.2	0.167	0.154	802	0.050	0.070	287	0.114	0.112	2478
TTM ≤ 2 M	0.099	0.109	4997	0.075	0.095	4087	0.103	0.103	27,615
2 M ≤ TTM < 6 M	0.103	0.100	6069	0.083	0.087	5004	0.145	0.115	28,618
TTM > 6 M	0.146	0.129	4886	0.099	0.084	3986	0.183	0.144	20,307

Panel B: RMSRE on implied volatilities of VIX options									
	In-sample			In-sample until Sep-08			Out-of-sample		
	w/o	w/	#	w/o	w/	#	w/o	w/	#
Overall	0.596	0.517	4768	0.533	0.422	3562	0.590	0.496	39,612
Mon. > 1.3	0.348	0.279	1312	0.086	0.048	4690	0.402	0.309	18,952
1.1 ≤ Mon. ≤ 1.3	0.362	0.268	1055	0.091	0.050	3120	0.443	0.318	6341
0.9 ≤ Mon. ≤ 1.1	0.527	0.383	1205	0.440	0.221	965	0.633	0.468	7325
Mon. < 0.9	0.943	0.877	1196	0.925	0.825	802	0.969	0.902	6994
TTM ≤ 2 M	0.584	0.518	2283	0.509	0.407	1662	0.533	0.425	6994
TTM > 2 M	0.586	0.432	2485	0.500	0.278	1900	0.612	0.522	11,355

Table 5

Estimated parameters of the following regression: $r_{t+h} = \alpha + \beta \times [\text{set of predictors}] + \epsilon_t$. The first two sets of predictors include the past RV, the model-implied one-month VRP (mVRP) and the slope of its term structure (VRP TS), measured as the three-month VRP minus the one-month VRP. Results are displayed for the benchmark model (bm) and for the SVJ3 model dynamics given respectively in Eqs. (11)-(12) and (11). The third set of predictors includes the age-related variance and the model-implied VRP only. The fourth set of predictors include the latent factors of the SVJ3 model. Finally, the last predictor is a model-free estimate of VRP (i.e., the difference between the realized return over the past 30 days and the current VIX value squared (RV-VIX²)). Before each estimate, we report its Hansen-Bushle's adjusted standard error.

	h = 1 month					h = 3 months					h = 5 months				
	bm	SVJ3	SVJ3	SVJ3	Model-free	bm	SVJ3	SVJ3	SVJ3	Model-free	bm	SVJ3	SVJ3	SVJ3	Model-free
constant	-0.015	0.007	-0.009	-0.050	-0.009	0.076	0.004	-0.009	0.343	-0.020	-0.107	-0.001	-0.030	-0.030	-0.023
	0.007	0.006	0.007	0.041	0.007	0.030	0.022	0.023	0.042	0.023	0.058	0.038	0.042	0.040	0.036
r				0.951					0.957					0.979	
				0.186					0.177					0.196	
n				-0.281					-0.662					-0.706	
				0.037					0.075					0.099	
s				0.059					-0.120					0.254	
				0.017					0.011					0.022	
RV	-1.005	-1.009	-0.722	-1.003		-2.001	-2.012	-1.538	-2.185		-2.878	-2.888	-2.136	-2.878	
	0.000	0.000	0.120	0.000		0.200	0.202	0.204	0.200		0.000	0.000	0.200	0.200	0.000
mVRP	-7.961	-12.036	-0.078			-17.779	-27.762	-13.274			-23.373	-37.367	-19.364		
	0.005	0.004	0.001			0.001	0.001	0.001			0.001	0.001	0.001		
VRP TS	13.986	9.161				20.872	20.254				37.021	25.078			
	1.643	1.580				0.043	0.200				0.000	0.000			
RV-VIX ²					-0.772				-1.841					-2.444	
					0.100				0.152					0.176	
Adj. R ²	0.022	0.023	0.001	0.020	0.007	0.009	0.007	0.010	0.076	0.007	0.015	0.016	0.031	0.060	0.011

	h = 6 months					h = 7 months					h = 8 months				
	bm	SVJ3	SVJ3	SVJ3	Model-free	bm	SVJ3	SVJ3	SVJ3	Model-free	bm	SVJ3	SVJ3	SVJ3	Model-free
constant	-0.020	0.004	-0.016	-0.018	-0.022	-0.196	0.007	-0.017	0.045	-0.019	-0.111	0.006	-0.009	-0.071	-0.013
	0.019	0.004	0.012	0.000	0.001	0.007	0.007	0.007	0.007	0.007	0.004	0.003	0.004	0.003	0.003
r				0.999					0.999					0.999	
				0.000					0.000					0.000	
n				-1.130					1.276					1.262	
				0.000					0.000					0.000	
s				0.000					1.200					1.079	
				0.000					0.000					0.000	
RV	-3.250	-3.250	-2.385			-3.430	-3.425	-2.343	-3.430		-3.285	-3.290	-2.120		
	1.007	1.000	0.502	1.000		1.200	1.200	0.470	1.200		1.200	1.200	0.400	1.000	1.000
mVRP	-20.127	-45.878	-21.550			-10.008	-46.700	-22.474			-10.008	-46.700	-22.474		
	0.000	0.000	0.000			0.000	0.000	0.000			0.000	0.000	0.000		
VRP TS	11.036	17.427	6.025			13.022	21.801	7.401			13.022	21.801	7.401		
	0.000	0.000	0.000			0.000	0.000	0.000			0.000	0.000	0.000		
RV-VIX ²	25.100	21.434			-2.648	23.244	25.503			-2.675	23.428	25.767			-3.501
					0.710				0.015					0.718	
Adj. R ²	0.012	0.012	0.016	0.014	0.018	0.006	0.006	0.006	0.047	0.018	0.021	0.022	0.001	0.021	0.000

5.6 Table 5: Return-predictability regressions

Read:

- Table 5 regresses future S&P 500 returns on model-implied VRP measures, realized variance, and latent factors.
- The VRP coefficient is negative because the VRP itself is usually negative: a larger absolute VRP predicts higher future returns.
- The VRP term-structure variable is positive, indicating that the shape of the VRP curve contains additional information.

Economic message. The VRP is not a one-dimensional object. Its level and term structure jointly encode information about future equity returns.

5.7 Table 6: Descriptive statistics

Read:

- Table 6 reports means, standard deviations, skewness, and kurtosis for S&P 500 returns, VIX returns, and squared VIX levels.
- The crisis-heavy in-sample period has especially high return kurtosis and high VIX variance.

Economic message. The sample contains extreme volatility episodes, making jump and stochastic-volatility features empirically necessary.

5.8 Table 7: Model selection

Read:

- Table 7 compares nested models using log-likelihood, AIC, and BIC.
- SVJ3 dominates the alternatives both without and with VIX options.

Table 6
Descriptive statistics for daily S&P 500 futures returns and daily VIX returns and square levels for the periods from March 2006 to February 2009 and March 2009 to April 2016. We report the mean (Mean), standard deviation (Std), skewness (Skew), and kurtosis (Kurt).

	March 2006–February 2009				March 2009–April 2016			
	Mean	Std	Skew	Kurt	Mean	Std	Skew	Kurt
S&P 500 ret.	−0.0009	0.0170	−0.1487	11.5442	0.0006	0.0107	−0.0879	7.4032
VIX ret.	0.0048	0.0774	1.3553	10.7419	0.0020	0.0750	1.2731	8.1333
VIX ²	0.0730	0.1038	2.7325	10.7644	0.0427	0.0341	2.2688	8.8821

more interesting is the comparison of pricing

Table 7
Log-likelihood (LL) and values of the AIC and BIC tests for subspec-

Table 7
Log-likelihood (LL) and values of the AIC and BIC tests for subspecifications of the SVJ3 model dynamics given in Eqs. (1)–(3) and for the benchmark model (dynamics given in Eq. (11)). The specifications considered are no jumps (SV2), m and u are constant (SVJ), u is constant (SVJ2), and no negative jumps in the variance (SVJ3 ^{$\Delta v > 0$}). The benchmark model has two factors and normally distributed jumps in the returns and their variance. The estimation period is from March 2006 to February 2009.

	Without VIX options					
	SV2	SVJ	SVJ2	SVJ3 ^{$\Delta v > 0$}	Benchmark	SVJ3
LL	9778	10,224	10,424	10,415	10,370	10,473
AIC	−19,538	−20,414	−20,802	−20,784	−20,704	−20,898
BIC	−19,496	−20,335	−20,695	−20,677	−20,621	−20,787

	With VIX options			
	SVJ2	SVJ3 ^{$\Delta v > 0$}	Benchmark	SVJ3
LL		9717	9729	9538
AIC		−19,388	−19,412	−19,040
BIC		−19,281	−19,305	−18,957

- Simpler models such as SV2, SVJ, SVJ2, and the benchmark model lose fit.

Economic message. The data require a flexible model with multiple volatility factors and jumps to describe the joint option markets.

5.9 Figure 2: Filtered volatility and jump trajectories

Read:

- Panel A plots the short-run variance factor v_t , which spikes during crisis periods.
- Panel B plots the persistent central-tendency factor m_t , which rises more gradually and decays slowly.
- Panel C plots the jump-intensity factor u_t , and Panel D shows filtered variance jumps.

Economic message. Volatility dynamics have both fast and slow components. Short-run variance captures sharp stress events; central tendency captures persistent volatility regimes.

5.10 Table 8: Diebold–Mariano pricing-error tests

Read:

- The Diebold–Mariano tests compare pricing errors from models estimated without versus with VIX options.
- Positive statistics indicate smaller pricing errors when VIX options are included.
- The VIX-option statistics are extremely large: **111.753** in sample and **133.872** out of sample overall.

Economic message. The improvement in VIX-option pricing is statistically overwhelming, not a small numerical artifact.

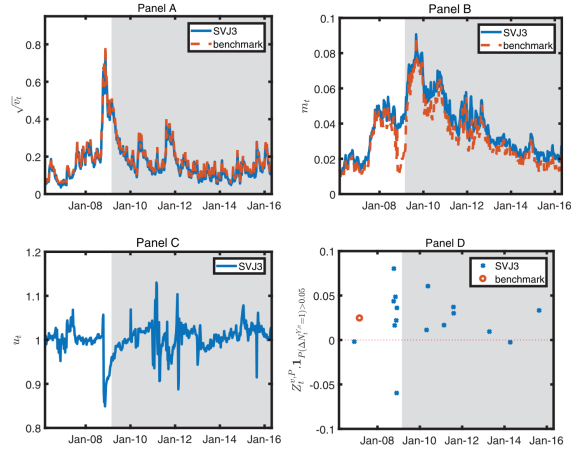


Fig. 2. Panels A and B plot the filtered trajectories of the latent volatility processes v and m for the SVJ3 and the benchmark model given in Eqs. (1)–(3) and (11), respectively, including VIX options in the estimation data set, from March 2006 to April 30, 2016. Panel C represents the filtered trajectory of the jump intensity factor λ , and Panel D displays the filtered size of jumps in the variance process v when the probability of jumps exceeds 5%. The shaded part of the graph represents the out-of-sample period, from March 1, 2009, until the end of April 2016.

Table 8

Statistics of Diebold–Mariano tests applied to option pricing errors. The pricing error at time t is defined as the mean square relative error between the model-implied option prices and observed prices. The test statistics compare the errors when excluding VIX options to the case when including VIX options. Under the null hypothesis that the two estimation data sets produce pricing errors of the same magnitude, the Diebold–Mariano test statistic is standard normal. A positive value means that the pricing errors become smaller when including VIX options.

	In-sample	Out-of-sample
S&P 500 options		
Overall	0.908	9.488
$Mon. < 0.7$	0.273	−6.252
$0.7 \leq Mon. \leq 0.95$	0.719	3.327
$0.95 \leq Mon. \leq 1.05$	−0.158	0.400
$1.05 \leq Mon. \leq 1.2$	0.000	0.693
$Mon. > 1.2$	−0.181	−0.167
$TTM \leq 2M$	−0.063	3.535
$2M \leq TTM < 6M$	0.338	2.712
$TTM > 6M$	0.422	5.721
VIX options		
Overall	111.753	133.872
$Mon. > 1.3$	7.432	26.020
$1.1 \leq Mon. \leq 1.3$	3.047	6.976
$0.9 \leq Mon. \leq 1.1$	3.130	15.870
$Mon. < 0.9$	17.423	132.233
$TTM \leq 2M$	13.062	23.411
$TTM > 2M$	9.928	120.845

5.11 Figure 3: Dynamics and decomposition of the variance risk premium

Read:

- Panel A shows that the six-month VRP is negative and drops sharply during the 2008 crisis.
- Panel B compares different maturities.
- Panels C–F decompose the VRP into continuous/jump components and into v_t and m_t contributions.

Economic message. Jumps matter strongly at short horizons, while the persistent variance factor matters for medium- and long-horizon VRP, especially outside crises.

5.12 Figure 4: Model-implied VRP versus model-free approximations

Read:

- Figure 4 compares the model-implied VRP to ex-ante and ex-post model-free approximations.
- The model-implied VRP is smoother because it uses the full state-space model rather than realized variance alone.

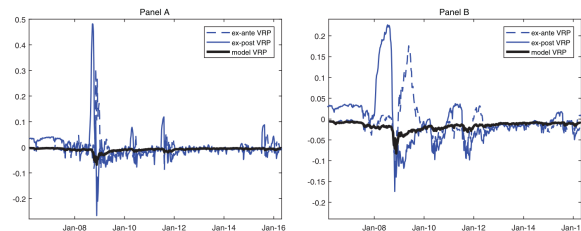
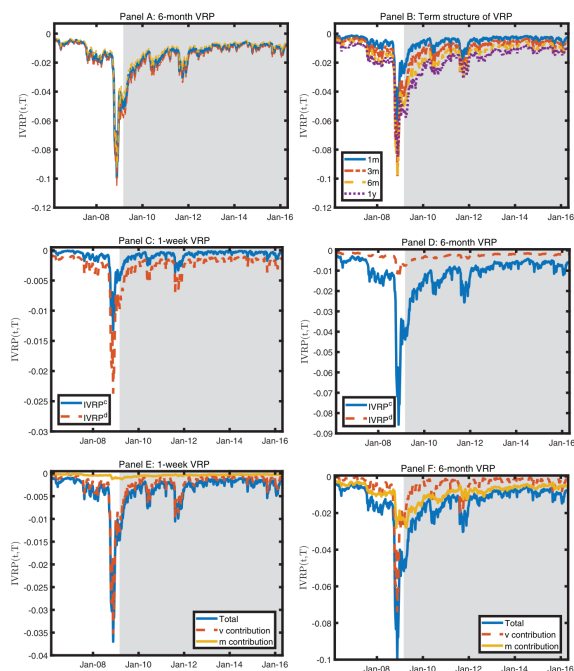


Fig. 4. Integrated VRP when estimating the SV3 model dynamics given in Figs. (1)–(3) using the full data set including VIX options (thick solid line), compared to the ex-ante model-free estimate of the VRP, computed as the difference between the past realized variance and the squared VIX (dashed line), and the approximated ex-post VRP, computed as the difference between the realized variance and the squared VIX (thin solid line). Panel A corresponds to an investment of one month, and Panel B to six months.

Economic message. Simple realized-variance proxies can be noisy. The model uses option prices and latent dynamics to infer a more structured premium.

5.13 Figure 5: Slope and components of the VRP term structure

Read:

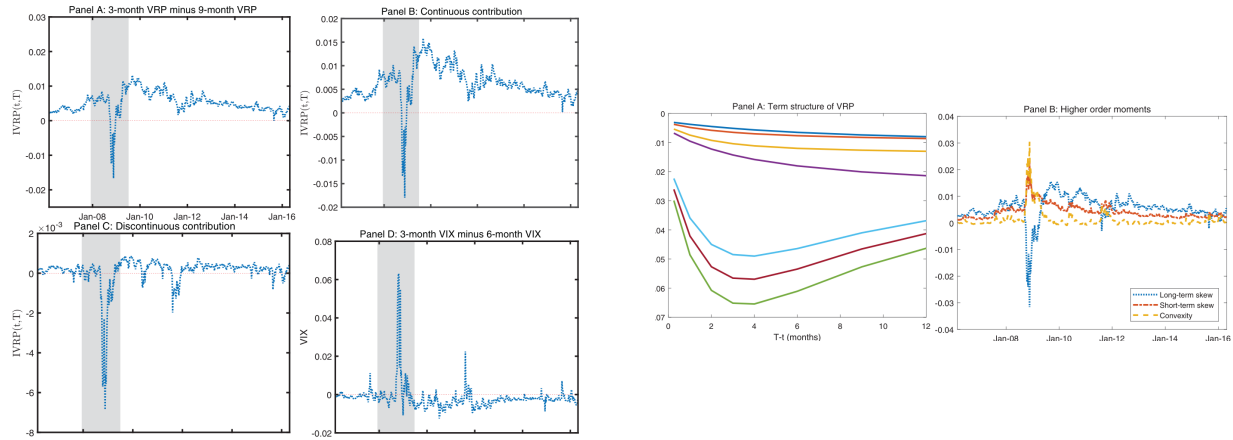
- Panel A plots the difference between the three-month and nine-month VRP.
- The difference usually has one sign but switches during the crisis.
- Panels B and C separate continuous and jump contributions, while Panel D compares synthetic VIX maturities.

Economic message. The slope of the VRP term structure changes sharply in stress periods. This provides the basis for the trading signal.

5.14 Figure 6: VRP term structure and higher-order shape

Read:

- Panel A shows the VRP term structure across maturities for selected dates.
- In calm periods, the term structure is generally monotone.
- During crisis periods, it becomes nonmonotonic, with a trough around three to four months.
- Panel B plots skewness and convexity measures of the term structure.



Economic message. Investors' willingness to insure against future volatility shocks changes with market conditions and is not limited to near-term realized volatility.

5.15 Figure 7: Forward-variance trading strategy

Read:

- Panel A shows weekly returns to a forward-variance investment.
- Panel B compares cumulative returns of the basic strategy and the strategies that pause or reverse exposure when the VRP slope switches sign.
- Panels C and D zoom in on the crisis period when the signal matters most.

Economic message. The VRP term-structure slope is economically useful because it identifies the weeks in which the standard variance-risk-premium trade suffers large losses.

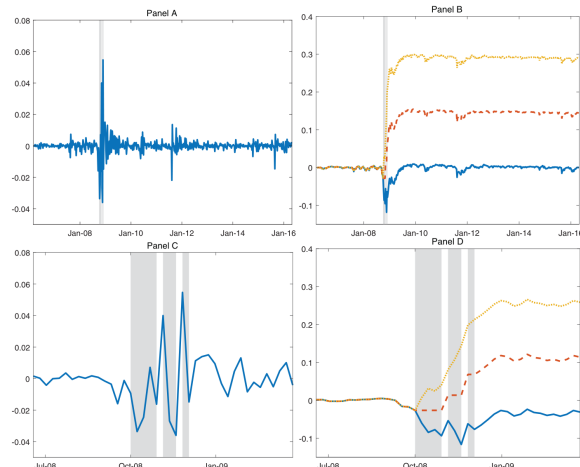


Table 9
RMSE on the synthesized squared VIX indices for maturities of two, three, five, and six months, excluding or including VIX options in the estimation data set. Results are reported over the entire time period (Overall), the in-sample period (IS), the in-sample period until September 2008 (IS S08), and the out-of-sample period (OOS). The indices are synthesized from S&P 500 options.

	w/o VIX options				w/ VIX options			
	Overall	IS	IS S08	OOS	Overall	IS	IS S08	OOS
SVJ3 model								
2m	0.018	0.017	0.008	0.018	0.015	0.013	0.007	0.015
3m	0.023	0.022	0.011	0.023	0.019	0.017	0.006	0.019
5m	0.029	0.028	0.015	0.030	0.024	0.021	0.007	0.024
6m	0.031	0.030	0.016	0.031	0.025	0.022	0.008	0.026
Benchmark model								
2m	0.017	0.018	0.016	0.015	0.014	0.014	0.007	0.014
3m	0.020	0.023	0.018	0.018	0.018	0.018	0.007	0.017
5m	0.025	0.028	0.022	0.021	0.022	0.022	0.010	0.022
6m	0.026	0.031	0.024	0.022	0.024	0.024	0.012	0.023

5.16 Table 9: Synthetic squared-VIX RMSEs

Read:

- Table 9 evaluates model pricing of synthetic squared-VIX indices for maturities of two, three, five, and six months.
- Including VIX options lowers RMSEs across maturities.
- In the SVJ3 model, the overall six-month RMSE falls from **0.031** to **0.025**.

Economic message. VIX options improve the estimated risk-neutral variance term structure, not just the fit of observed VIX-option prices.

6 Short summary

- The paper asks whether VIX options contain information about S&P 500 volatility dynamics beyond S&P 500 options.
- It estimates a flexible affine jump-diffusion model with two variance factors and a jump-intensity factor.
- Estimation uses an auxiliary particle filter and a large panel of S&P 500 and VIX derivatives.
- VIX options are not spanned by S&P 500 options: including them materially improves VIX-option pricing in and out of sample.
- The full SVJ3 model is preferred to simpler nested alternatives.
- Including VIX options improves the model's representation of the variance term structure.
- The variance risk premium is negative and highly sensitive to market stress.
- The VRP term structure is monotone in calm periods but can become nonmonotonic during crises.
- Jumps dominate short-horizon VRP behavior; persistent variance dominates medium- and long-horizon VRP behavior outside crises.
- The VRP term-structure slope provides a trading signal and helps forecast S&P 500 returns.

7 Conclusion

7.1 Core conclusion

VIX options contain valuable and economically meaningful information about the risk-neutral distribution of future volatility. This information is not fully spanned by S&P 500 options. A joint model that includes VIX options better prices VIX derivatives, better reproduces the variance term structure, and produces more useful variance risk premium estimates.

7.2 Economic intuition

S&P 500 options price the future distribution of equity returns. VIX options price the future distribution of volatility. Since volatility itself is stochastic, persistent, and jump-sensitive, a claim on future volatility contains information that cannot be reduced to a single current VIX level or to S&P 500 option prices alone.

7.3 Methodological takeaway

The paper shows how to combine affine option pricing, Fourier-cosine methods, and particle filtering to estimate a high-dimensional option-pricing model using the full cross-section and time series of option prices.

7.4 Asset-pricing takeaway

The variance risk premium is a term-structure object. Its level, slope, convexity, and decomposition across variance factors matter for volatility trading and expected stock returns.

7.5 Final takeaway

For empirical asset pricing and derivatives research, VIX options should not be treated as redundant. They are a distinct source of information about volatility risk, especially about the conditional distribution of future variance and the price investors place on volatility shocks.